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Situation awareness for engine room monitoring at future shore control centre

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ABSTRACT

Situation Awareness (SA) is a cornerstone for system verification and validation, particularly vital at the future Shore Control Centre (SCC). However, engine room monitoring (ERM) receives little attention regarding the SA. This research conducted two Likert-type surveys with 201 marine engineers and employed a descriptive statistic and the Mann–Whitney U test to investigate cue strengths in forming comprehension around the engine room and generality in cue strengths by ship types, experience and ranks of respondents. The importance varies to different extents, and their perspectives seem not to deviate much, with the max disagreement rate at 12% by ship types and the ranks of respondents. The extent of disagreement tends to reduce with the increment of experience thresholds. The outcomes offer a promising strategy for developing a human-centred monitoring system to mitigate the potential risk of information overload and incorrect schema instantiation at the future SCC.

KEYWORDS

Engine room monitoring; shore control centre; situation awareness; cue strength

1. Introduction

Increasing interest in the economic benefits of autonomous vessels has introduced remote operation as a transient step to achieve full autonomy, representing the second and third degree of autonomy (IMO 2024). However, remote operation accompanies the removal or replication of ship sense, e.g. vibration, visibility and body orientation, which traditionally play a vital role in acknowledging environmental cues that form SA. For instance, Yoshida et al. (2020) identified that visibility was involved in accessing most perceptual information, followed by body balance, thermal sense, sound, and smell during navigation. An emphasis on SA for remote operation was also placed in the two autonomous ship feasibility projects, concluding that a lack of ship sense in the SCC would be a critical challenge for operators to understand the situation and the absence of full comprehension of the ship's condition could lead to incidents (AAWA 2016; MacKinnon, Man, and Michael 2015).

These lessons have been incorporated into contemporary industrial initiatives and the research sector. For example, the Korea Autonomous Surface Ship (KASS) project entailed

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the SA support system as a part of an intelligent navigation system, e.g. object detection (KASS 2021). Similarly, the Designing the Future of Fully Autonomous Ship (DFFAS) project developed the 3D surrounding bird view to assist remote operators (NYKLINE 2020). In the research sector, there is an emerging interest in revealing the information requirements through goal-directed task analysis according to the three levels of SA theory: perception, comprehension and projection (Endsley et al. 1998). These endeavours include SA information requirements for ship pilotage (Sharma, Nazir, and Ernstsen 2019) and dry docking (Haffaci, Massicotte, and Doyon-Poulin 2021).

The present issue with the current paradigm lies in conceiving a vessel as merely a vehicle, leading to excessive emphasis on remote navigation and complete negligence on remote ERM. This narrow conception appears to result from employing the SA from the aviation and automobile sectors into the maritime industry without ruminating industrial differences. Remote aerial vehicles are much smaller, less complicated, and run for shorter periods, so the chances of technical malfunction when in action are minimal. This allows operators to be attentive in maneuvering under a volatile environment, not in detecting irregular changes in the system state for future inspection or maintenance. The latter, however, becomes vital for ocean-going vessels where monitoring and diagnosis activities on numerous systems and machinery are undertaken at sea and port to ensure operational continuity.

The marine engine room is similar to the power plant control rooms in that both are embedded with numerous systems and how awareness of the systems is developed and vital for operation safety. Similar to the engine room, the active exploration of information through review logs, shift turnover, panel walkthroughs, and field tours is central to developing the SA over the plants (Lau et al. 2011). Otherwise, it could lead to the incidents. For instance, Patrick, James, and Ahmed (2007) investigated where SA was lost in forty-one nuclear power plant incidents. The findings identified that deficiency of awareness on an action performed or omitted was the most prevalent, accounting for 19 incidents. A lack of awareness of plant and sub-system status was found in 18 incidents. They also noted that over half of the incidents involved attention deficit from distraction and multi-tasking, causing the loss of SA. Similarly, a more recent study examined 58 incidents during the planned outage of nuclear power plants (Solberg, Nystad, and McDonald 2023). They discovered that over half of incidents involved comprehension errors such as lack of mental model and over-reliance on default values. This suggests that the SA support systems must be approached from maneuvering and system monitoring perspectives, entailing both remote navigation and ERM.

Task analysis is one of the common approaches to revealing context-specific information, serving as an initial step in designing the supporting interface. In the context of nuclear power plant monitoring, the ecological interface design developed from the domain task analysis was found to help operators respond to unexpected events better than traditional display (Burns et al. 2008; Kim et al. 2012). Similarly, the task analysis was employed to investigate critical cues for electric grid monitoring (Stevens-Adams et al. 2015). For SA, the goal-directed task analysis has been employed to identify the information requirement for each objective in multiple areas, including fire response teams (Rezaeifam, Ergen, and Günaydın 2023), brigade operation (Bolstad et al. 2002) and recently for ERM (Han et al. 2024). These requirements can further be used as transparent information that describes a rationale behind the machine agent's analysis to keep operators aware of their intention (Trujillo et al. 2015). The endeavour to integrate

transparency into Endsley's SA model was made in the work of (Chen et al. 2014), postulating the Situation Awareness-Based Agent Transparency (SAT) with an emphasis on human and machine interaction. The SAT model has been employed primarily in the aviation and military domains, and positive associations with trust, performance and response time were reported (Han et al. 2021).

However, the goal-directed task analysis and its information requirements are limited because every information element is treated equally, albeit some cues might be more critical than others in understanding the situation. In other words, perception information elements (e.g. simple readouts) may not contribute equally to forming mental models or schemata of comprehension information (e.g. combustion status), and few critical elements (e.g. max cylinder pressure) might be sufficient to reflect the environment. Mutzenich et al. (2021) made a similar assertion that vehicle colour may not be as important as vehicle trajectory for remote drivers, and a minimum SA requirement would be critical for future training. Similarly, Endsley (2016) explains that schema is activated through pattern-matching between environmental cues and memory, and activation does not require all information comprising schemata. Schema activation was further investigated in terms of the text understanding. For instance, Anderson et al. (1976) found that instantiation-relevant cues reminded participants better of the sentences they read than general cues. The instantiation-relevant cues were the images of the subject (i.g., actress) in the given sentence 'the woman was outstanding in the theatre', and general cues were the subject (i.g., woman). Similarly, (An 2013) suggested that schema instantiation does not require all memory markers when a cue is highly indicative, although it may instantiate multiple schemata that share the cue.

The above schema characteristics present a new approach to enhancing SA under a high risk of information overload, envisaged to be severe due to the replication of ship sense and duplicated usage of perception elements for higher-level SA information Han et al. (2024). The latter escalates the quantity of information at the expense of user understandability and presents a great challenge in integrating system transparency and SA for complicated systems with numerous information elements. In this respect, cue strength differences in schema instantiation can be promising in alleviating the interface complexity and information overload of a large-scale socio-technical system while still providing a similar or the same level of SA with less information. The range of available information can be adaptable to prevent the activation of a wrong schema that shares important cues with a correct schema and to minimise cognitive load for processing information under volatile environments, e.g. congested water, allowing operators to allocate their mental resources to external stimuli. On the contrary, the range can be extended to simulate remote operators under monotonic environments, e.g. calm water, and resolve passive fatigues from tedious monitoring. However, no efforts have been made to investigate the cue strength of perception information in constructing comprehension over the ship engine room. Therefore, this paper examines the cue strength differences through quantitative surveys with 201 marine engineers based on the SA information requirements developed in the previous research (Han et al. 2024).

2. Materials and methods

This research has been approved by the University of Tasmania Human Research Ethics Committee (Project ID: 24900). Two Likert-type questionnaire surveys were designed to

identify the importance of perception elements in forming comprehension elements for ERM. Participants were directed to provide their current or the latest ranks, the vessel type in which they spent the most time, and their total seafaring experience with off-time included. The questions were developed and derived from the SA information requirement for ERM (Han et al. 2024). The five-point importance rating from not important to very important was used to minimise the time demand while providing good point separation. The selection of the machinery for each questionnaire was based on the common practice that a group of machinery is allocated to individual engineers as the main responsibility, such as the main engine for a first-assistant engineer and auxiliary boiler for a third-assistant engineer. For data analysis, the Mann–Whitney U test, one of the non-parametric tests suitable for ordinal and not normally distributed data and two-group comparisons, was used to compare participant responses by ship type, experience, and rank.

2.1. Questionnaire A

Questionnaire A contains 124 questions, 32 comprehension information, and four types of machinery: main engine, generator, air compressor, and steering gear. Each question inquires about the importance of a particular perception element in evaluating matching comprehension information. The questionnaire is distributed to former or current senior engineers (chief or first-assistant engineers).

2.2. Questionnaire B

Questionnaire B contains 115 questions, 34 comprehension information, and five types of machinery: auxiliary boiler, freshwater generator, purifier, oily water separator, and air handling unit. The questionnaire is distributed primarily to former or current junior engineers (second-assistant and third-assistant engineers). However, senior engineers can participate in Questionnaire B, as their experience is accumulated from lower ranks, and their responsibility is typical across the entire engine room.

2.3. Participants

The participants were recruited through the researcher's professional contacts from August 2022 to March 2023. As a result, two hundred-one marine engineers participated in the online surveys through Google Forms. The sample numbers for each questionnaire were above the minimum sample size of 96 for an unknown population, calculated from the following equation. (1). Several maritime qualitative studies used a similar approach (Islam et al. 2017; Islam et al. 2018).

$$\text{Minimum sample size } n = \frac{Z^2 P(1 - P)}{e^2} \quad (1)$$

Z corresponds to 95% confidence of 1.96; e is the margin of error ($e = \pm 0.10$); P equals the level of satisfaction with a median of 0.50. The participant details are shown in Tables 1 and 2.

Table 1. The participant details on each questionnaire.

Engineer Ranks	Questionnaire A	Questionnaire B
	Number (N) & Average Experience Years	
Chief Engineer	N = 32 with 21.3 years	N = 4 with 14.8 years
First-Assistant engineer	N = 68 with 6.9 years	N = 25 with 6.0 years
Second-Assistant engineer	–	N = 61 with 4.8 year
Third-Assistant engineer	–	N = 11 with 3.0 years
Total	N = 100 with 11.5 years	N = 101 with 5.28 years

3. Results

3.1. Central transparency

The medians on each information element are calculated for central transparency (Joshi et al. 2015). The survey responses on the generator are described in Figure 1, and the analysis of other engine room components can be found in Appendix A. Each boxplot indicates a comprehension element named an alphabet with numbers; its perception elements are on the x-axis. Further descriptions of the information elements can be found in Appendix B. The medians for every perception element are at least important [4], and the same trend is observed in other machinery groups, validating the SA information requirements of (Han et al. 2024). The majority of responses in B3.2 is important [4, $n = 63$], while twenty-two responses are very important, and twenty-six responses are

Table 2. The participant details on ship types.

Group	Sub-Group	Ship Type	Number
Bulk Carrier	Tanker	Very Large Crude Carrier	7
		Mid Range Tanker	1
		Oil Tanker	11
		Tanker	7
		Chemical Tanker	12
		Petroleum Tanker	1
		Product Tanker	2
		Shuttle Tanker	2
		LPG	12
		LNG	27
	Gas Carrier	Gas Carrier	1
		Bulk Carrier	26
		Coal/Ore Carrier	4
		Pure car, truck carrier (PCTC)	7
General Cargo	Car Carrier	Car carrier	12
		Ro-Ro	2
		Container	40
		Heavylift	4
	General Cargo	General Cargo	8
		Passenger Ship	2
	Supply Vessel	RO-Pax	2
		Offshore Support Vessel	2
		Supply	2
		Platform Supply Vessel	1
Others	Drilling Rig	Drilling Rig	1
		Training Ship	1
		Tug	1
		Warship	1
	Non-Identifiable	Non-Identifiable	2
		The total number of participants	201

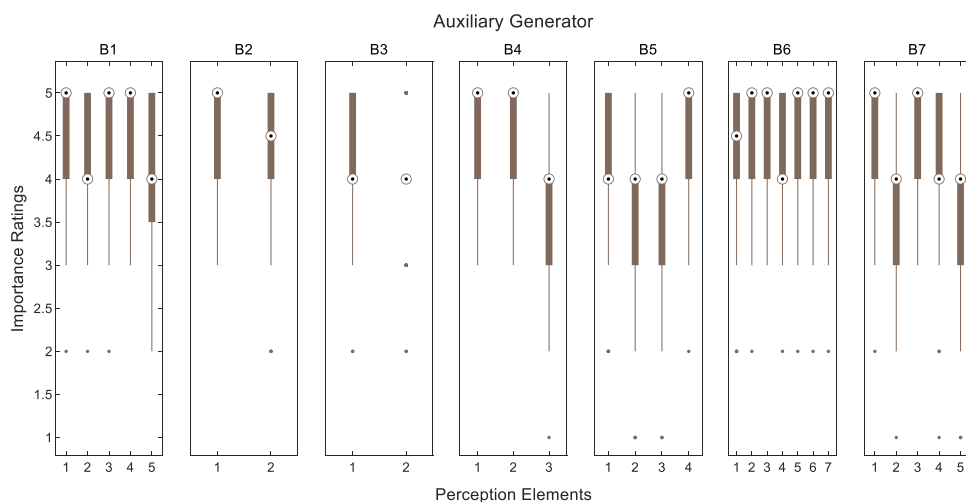


Figure 1. The survey responses on the auxiliary generator. Outliers and medians are shown as filled circles and a black dot inside a white circle, respectively.

either moderately important [3, $n = 13$] or slightly important [2, $n = 13$], resulting in the same percentiles for 25th and 75th.

3.2. Cue strength generality for questionnaire A

The career profile of marine engineers is typically represented by vessel type, sea experience, and occupational rank, which can influence their importance ratings on each perception element. Hence, the study investigates the generality of survey responses by three categories: (a) ship types by general cargo and bulk carrier, (b) ranks and (c) sea experience above and equal to or below a threshold. The general cargo group includes car carriers, containers and heavy lifts, while tankers, gas, and bulk carriers are included in the bulk carrier group. Other ship types and non-identifiable samples are excluded from ship type comparisons. Using a certain year threshold to differentiate experience groups may yield a substantially different trend; therefore, different thresholds with 0.5-year increments are used. The ship type and rank comparisons are shown in Table 3.

Eight samples belonging to neither bulk nor general cargo ships are excluded, leaving 91 samples for ship-type comparison. The alphabet abbreviations indicate the machinery name, followed by an integer for comprehension information and a decimal for perception information. For instance, A5.1 represents the importance of the turbocharger RPM (1) on the adequacy of the turbocharger (5) for the main engine (A). A list of SA information elements and their ratings can be found in Appendix A. The analysis indicates significant differences in responses to six perception elements which account for around 5% of the total number of questions ($n = 126$). The mean ranks show that the bulk group considered given elements more important than the general cargo group. However, it should be noted that these differences are relative only, as the medians of the general cargo group for all elements are at least important [4]. All samples ($n = 100$) are used for rank-type comparison, and a similar trend is observed. The importance ratings on the

Table 3. The Mann-Whitney U test for the ship type and rank groups for Questionnaire A.

By Ship Type	A5.1		A5.3		B5.1		B5.4		B7.2		C4.1	
Mann-Whitney U	631.500		672.000		645.500		703.000		707.500		711.000	
Exact Sig. (2-tailed)	0.002		0.014		0.007		0.025		0.036		0.040	
Group (Sample Number)	Bulk (59)	Cargo (32)	Bulk (59)	Cargo (32)	Bulk (59)	Cargo (32)	Bulk (59)	Cargo (32)	Bulk (59)	Cargo (32)	Bulk (59)	Cargo (32)
Mean Rank	51.30	36.23	50.61	37.50	51.06	36.67	50.08	38.47	50.01	38.61	49.95	38.72
Median	5	4	4	4	5	4	5	4	4	4	4	4
By Rank	A2.4		A6.1		A12.1		A12.2		C4.2		C6.1	
Mann-Whitney U	783.500		720.000		801.000		791.000		837.500		832.000	
Exact Sig. (2-tailed)	0.015		0.003		0.021		0.014		0.048		0.044	
Group (Sample Number)	Chief (32)	First (68)	Chief (32)	First (68)	Chief (32)	First (68)	Chief (32)	First (68)	Chief (32)	First (68)	Chief (32)	First (68)
Mean Rank	60.02	46.02	62.00	45.09	59.47	46.28	59.78	46.13	58.33	46.82	58.50	46.74
Median	5	4	5	4	5	4	5	4	5	4	4	4

six elements are significantly different. Chief engineers rate these elements more important than first-assistant engineers, whose medians for all elements are at least important [4].

The participants are categorized into two groups: less experienced and more experienced in Figure 2. The survey group is divided by experience thresholds with 0.5 year increments from 4 years to 14 years. The sample number for each subgroup is described as a bar in which the lower body represents the number below the threshold while the upper body indicates the number over and equal to the threshold. When the participant group are divided by the 6-year threshold, a number of perception elements with a significant difference are at a peak ($n = 48$), accounting for over 38%. The 32 samples belong to the sub-group below the threshold year, while another group has 68 samples. With the threshold of 6.5 years, nine samples with experience between six and six point five years are added to the less experienced group. The extent of statistically significant differences reduces by up to 8%, that is ten perception elements. Interestingly, it becomes smaller with an increasing year threshold from the 7-year mark, and the importance rating on only two perception elements is significantly different at the 14-year mark.

3.3. Cue strength generality for questionnaire B

The ship-type comparison includes two groups composed of ninety-five samples, 54 and 41 samples for bulk and cargo groups, respectively. Six samples are excluded, belonging

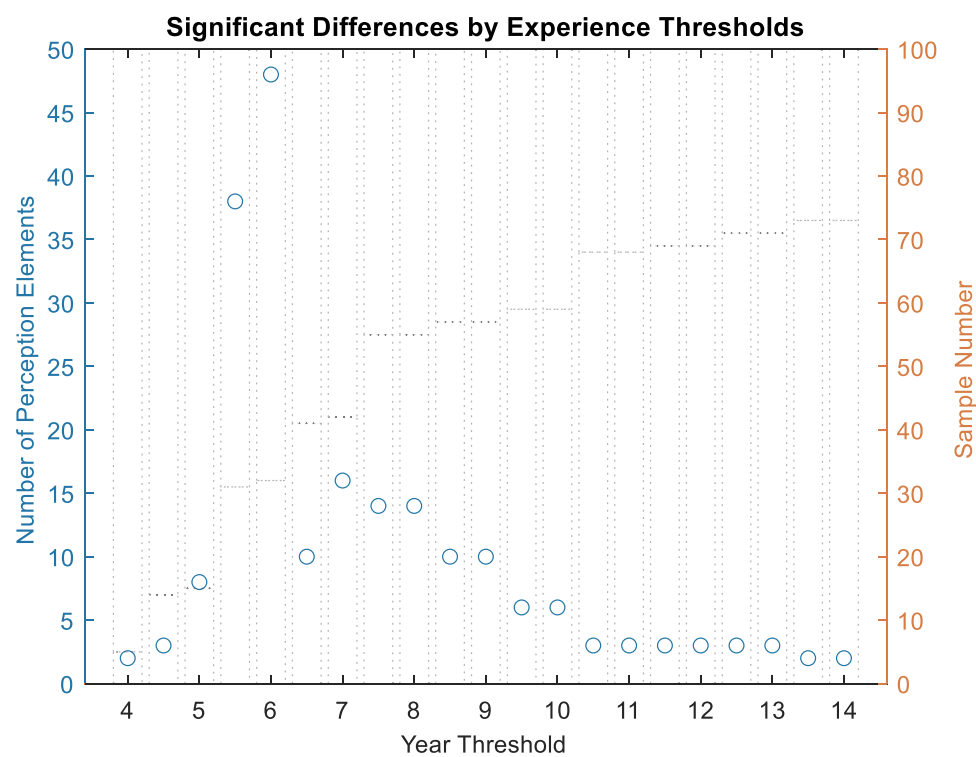


Figure 2. A number of perception elements with a significant difference.

to other vessel types or are non-identifiable, which participants respond as diesel or all instead of a particular ship type. The result demonstrates significant differences in their importance ratings on 14 perception elements, accounting for 12% of the total number of questions ($n = 115$). Mean ranks reflect higher importance ratings on these elements by the general cargo group than the bulk group. The differences in median values are found in three elements in which the bulk group rate as moderately important [3] lower than that of the cargo group at important [4]. Four rank-type comparisons employ only 100 samples due to the partly missing data by one junior participant, such as E7.4 and E8.2. Other than that, all samples ($n = 101$) are employed for the analysis. The result shows significant differences exist in eleven perception elements, accounting for around 10%. Interestingly, the junior group rates those higher than the senior group, as shown by mean ranks. There is a difference in median values on the two perception elements, which the junior group rated as important [4] higher than that of the senior group at moderately important [3], shown in [Table 4](#).

Likewise, the participants are categorized into two groups: less experienced and more experienced shown in [Figure 3](#). The survey group is divided by experience thresholds with 0.5 year increments from 1.5 years to 10 years. The sample number for each subgroup is described as a bar in which the lower body represents the number below the threshold while the upper body indicates the number over and equal to the threshold. The biggest difference in the number of elements ($n = 34$) is observed with the 1.5-year threshold, which makes up around 30%. The sample sizes of both groups in comparison are 10 and 91 for less experienced and more experienced groups, respectively. The inversely proportional relationship between the number of perception elements with a significant difference and the year threshold is observed. Between the 3.5-year and the 9-year thresholds, the number difference in percentile reduces from 5% to 0%. Above the 9-year threshold, no statistically significant difference is found in both groups' importance ratings on the perception elements.

4. Discussion

4.1. Representative strengths of perception information

This study discovers that strength differences exist in perception elements comprising comprehension around engine room machinery. Median values indicate that 71.5% of all elements are considered important, while 26.8% are rated very important. Only one element, H4.5: bilge temperature, is regarded as moderately important in comprehending the adequacy of the bilge holding tank. The rest lies between important and very important, with a median of 4.5, accounting for 1.3% of all elements. These findings offer a novel human-centred approach to resolving a potential information overload when designing the support system for a large-scale system embedded with many interrelated machinery and systems striving for operators' attention, and their ongoing awareness is critical for responding to unexpected events or automation failures. Its central concept lies in the fact that not all information is required to activate a mental model of an object, text, or situation (i.g., categorisation mapping), and particular cues activate better than others (i.g., schema instantiation). That means acquiring the same or similar level of awareness with less but important cues might be possible. For example, power generation status

Table 4. The Mann-Whitney U test for the ship type and rank groups for Questionnaire B.

By Ship Type	E6.3	E7.2	E7.3	E8.2	E9.3	F3.1
Mann-Whitney U	830.000	790.500	817.500	849.000	802.500	836.500
Exact Sig. (2-tailed)	0.022	0.017	0.034	0.050	0.022	0.040
Group (Sample Number)	Bulk (54) Cargo (41)	Bulk (54) Cargo (41)	Bulk (54) Cargo (41)	Bulk (54) Cargo (41)	Bulk (54) Cargo (41)	Bulk (54) Cargo (41)
Mean Rank	42.87	54.76	54.74	54.06	53.28	54.44
Median	4	4	3	4	4	4
	F5.3	F6.3	G6.2	G6.3	H4.5	I3.2
Mann-Whitney U	784.500	834.500	758.000	670.000	841.000	775.000
Exact Sig. (2-tailed)	0.009	0.031	0.005	0.001	0.039	0.009
Group (Sample Number)	Bulk (54) Cargo (41)	Bulk (54) Cargo (41)	Bulk (54) Cargo (41)	Bulk (54) Cargo (41)	Bulk (54) Cargo (41)	Bulk (54) Cargo (41)
Mean Rank	42.03	55.87	42.95	56.51	43.07	54.49
Median	4	4	4	3	3	3
	I4.4	I4.5				
Mann-Whitney U	707.000	814.500				
Exact Sig. (2-tailed)	0.001	0.020				
Group (Sample Number)	Bulk (54) Cargo (41)	Bulk (54) Cargo (41)				
Mean Rank	40.59	57.76	42.58	55.13		
Median	4	4	4	4		
By Rank	E7.4	E8.2	E9.3	G5.3	G6.4	H1.2
Mann-Whitney U	696.500	668.500	759.000	769.500	708.000	707.500
Exact Sig. (2-tailed)	0.006	0.002	0.024	0.027	0.005	0.006
Group (Sample Number)	Junior (71) Senior (29)	Junior (71) Senior (29)	Junior (71) Senior (29)	Junior (72) Senior (29)	Junior (72) Senior (29)	Junior (72) Senior (29)
Mean Rank	55.19	39.02	55.58	38.05	54.31	41.17
Median	4	4	4	4	4	4
	H1.3	H2.3	H4.4	H5.1	H5.4	H5.4
Mann-Whitney U	725.000	789.500	644.000	738.000	727.500	
Exact Sig. (2-tailed)	0.013	0.046	0.001	0.016	0.012	
Group (Sample Number)	Junior (71) Senior (29)	Junior (72) Senior (29)	Junior (72) Senior (29)	Junior (72) Senior (29)	Junior (72) Senior (29)	
Mean Rank	54.79	40.00	54.53	42.22	56.56	37.21
Median	4	4	4	3	4	4

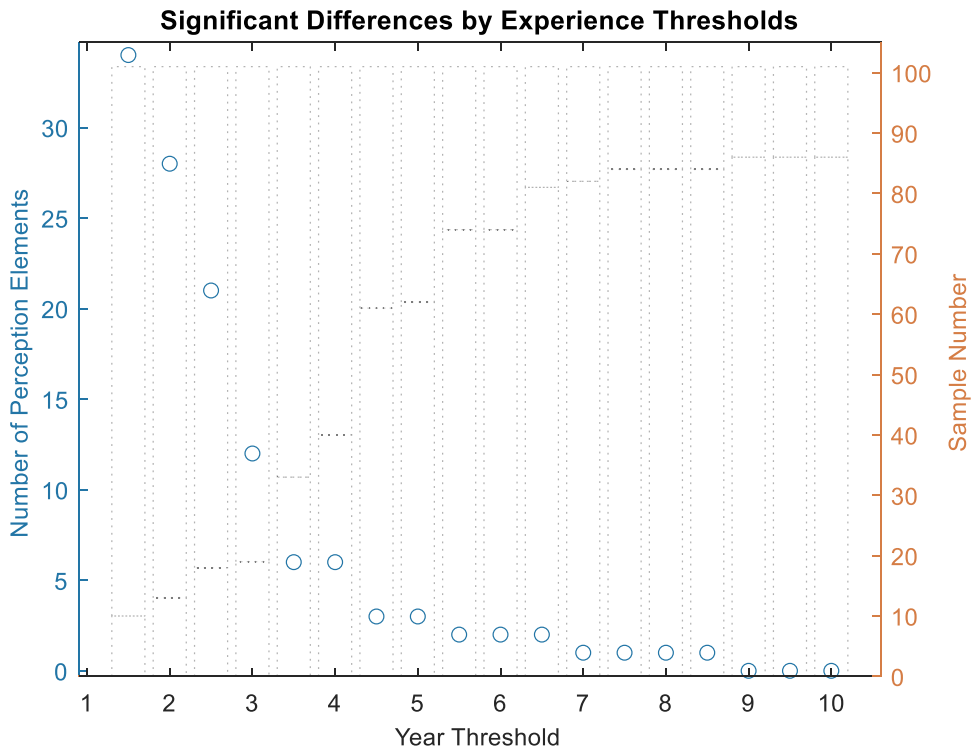


Figure 3. A number of perception elements with a significant difference.

is a critical facet in monitoring the ship’s main engine and is reflected through four perception elements. Of the four elements, participants reported that the engine revolution per minute was the most important, while brake power and target revolution were the least important. This implies a great potential for comprehending power generation status with engine revolution per minute information only, which is the strongest feature of the power generation status schema. The application can be further extended to mitigate the overload of transparent information that emerged from the duplicated usage of a single piece of information (e.g. engine speed) in describing multiple phenomena (e.g. power generation status and fuel consumption rate).

However, a major hindrance from presenting key perception elements to mitigate information overload is the possibility of distortion in meaning because the schema provides default values for missing information. Anderson et al. (1977) explained that it occurs when incomprehensive writing requires readers to fill in the missing information. Misinterpretation is more likely when key markers are missing or are shared by multiple schemata without additional descriptors to specify them. The oil leakage accident in the Kalamazoo River in 2010 represents the latter case, although it was considered a failure to attend to repeated warnings due to automation bias and complacency in their report (Wesley and Dau 2017). The low-pressure alarm was triggered during shutdowns, and the engineer viewed this as column separations, harmless and common in the shutdown process, not the pipeline leakage. The low-pressure alarm could fit into both column separation and pipeline leakage schema, and no further descriptors were present to

Table 5. The key markers and additional descriptors for the auxiliary generator.

Comprehension	B2. Status of Generator		B5. Adequacy of Fuel Consumption Rate	
	Load			
Key markers	B2.1 Generator load	5	B5.4 Generator load	5
Additional Descriptors	B2.2 Generator capacity	4	B5.1 Daily fuel consumption rate	4
			B5.2 Average daily fuel consumption rate during past voyage	4
			B5.3 Average daily fuel consumption rate during sea trial	4

distinguish them. This shows that the accident may not be the result of complacent behaviour and automation bias, but it is due to the poor understanding of how one comprehends the situation in designing the monitoring system.

The extent of available information on the screen can be adjusted to prevent incorrect schema instantiation and distinguish similar schemata that share the same cues. For instance, the SA support system can display key markers to avoid information overload based on cue strengths and provide additional descriptors if there are more than two comprehension elements with the common key marker to avoid incorrect understanding. Table 5 describes an example in which a median of five is used to differentiate between key markers and additional descriptors for the auxiliary generator. The generator load (key marker and perception element) is considered the most important perception element in comprehending the status of the generator load and the adequacy of the fuel consumption rate (comprehension elements). The key marker helps remote operators develop a swift understanding of these facets, but in the event of a system malfunction, the load information is not comprehensive enough to reveal the precise cause without additional descriptors. In this regard, the capability to adjust the range of information promises a shortcut for acquiring SA and provides a comprehension description for diagnostic activities. Such a filtering method might also be employed to regulate cognitive load for information processing, considering the operating environments that demand different levels of mental resources, such as calm and congested water.

4.2. Generality in cue strength

This study investigates the generality of importance ratings on each questionnaire by three categories: vessel type, sea experience, and occupational rank. In Questionnaire A, when the group is divided by vessel type and occupational rank, a significant difference in their ratings is observed in only six perception elements, accounting for around 5% of all perception elements. This indicates that a similar perspective over perception elements among senior engineers exists regardless of their vessel types and occupational rank. The finding can be attributed to the fact that embarking on different types of ships throughout their careers is common for marine engineers. They could have worked in a diesel-driven vessel, which both questionnaires are based on, or have been exposed to many noble situations in the engine room. Reuber (1997) explains that expertise is achieved through novel situations instead of repetitiveness. In addition, senior engineers typically hold long experience, including years as a junior which may contribute to knowledge saturation. The 10,000-hour rule explains that it takes around 10,000 hours to achieve an expert level of skills (Ericsson, Krampe, and Tesch-Römer 1993). Regardless of many controversies around its validity, due to its simple time-expertise relationship, the sea

experience to become the first engineer may have been sufficient to develop proper cue strengths in their schemata similar to those of chief engineers.

When the group is divided by experience from 4 to 14-year thresholds with 0.5 years increments, the number of perception elements having significant differences tends to reduce with increasing year threshold. With the threshold of 14 and 4.5 years, the numbers of perception elements are the second lowest, while the lowest number is observed with the threshold of 4 years. The small group difference at 4, 4.5, and 5-year marks is likely because one group has much smaller samples, which are 5, 14, and 15 versus 95, 86, and 85 samples, respectively. The year thresholds for groups having at least 30 samples in each subgroup are from 5.5 to 12 years. One implication for future SA studies for ERM is that it may not be essential, although desirable for higher statistical power, to recruit a large number of highly experienced engineers, as there is not much difference between the chief and first engineers' perspectives on the importance rating when their experience exceeds around 6.5 years. Their perspectives become almost identical at 12 year-mark, with only three perception elements having a significant difference, accounting for 2%. A smaller number of highly experienced engineers or a larger number of less experienced engineers can be involved to understand the SA during ERM, particularly cue strengths of perception elements withdrawn from systems typically under their responsibility.

A different trend is observed in Questionnaire B when the group is divided by vessel type and occupational rank. The vessel-type and occupational rank comparisons show a significant difference in 14 perception elements out of 114 elements (12%) and 11 perception elements (10%), respectively. While these differences are not substantially large, the differences are higher than that of Questionnaire A. It may be because most respondents are junior engineers, over 70% and, therefore, have limited experience with various types of vessels. The narrow experience with a particular type of vessel, such as a liquid bulk carrier, can prevent them from accurately evaluating the perception elements of diesel-driven vessel machinery due to the differences in the engine room type and machinery. On the other hand, most junior engineers are third engineers (61 out of 101) with an average experience of 4.8 years, and machinery items on Questionnaire B are generally under their responsibility. These could alleviate perspective differences from being further noticeably dissimilar.

The perspective differences by experience thresholds show a declining trend along with threshold increments. No difference is observed between a group with over ten years of experience and the rest, while the comparison with a 1.5-year threshold indicates a significant difference on 34 perception elements (30%). The limited experience over a wide range of vessels and insufficient time to develop adequate cue strengths likely contribute to the substantial viewpoint difference in the latter group. The fact that only six perception elements (5%) have a significant perspective difference with the 3.5-year threshold proposes a similar suggestion for conducting future ERM SA studies regarding sample group size and experience.

Most importantly, the high similarity in cue strengths regardless of vessel type suggests that the current findings can serve as a prototype layout in the design spiral of future remote monitoring stations. The layout can be enhanced further by incorporating specific machinery into the information requirement for a particular vessel type, such as a crane for containers and a hatch cover for bulk carriers.

5. Limitation

One of the limitations involves the uncertainty of evaluation criteria for importance ratings. The criteria may have included the range of monitored events, the frequency of occurrence, and the criticality of the event that a particular element represents. In other words, one may consider it important because it represents (a) the most monitoring events, (b) the most common event, and (c) the most critical event. If the importance rating were primarily based on the criticality of the event, the usage of key markers would be limited to those critical and rare events, not representing the common and a wider range of events during the ERM. However, the current study provides a novel human-centred approach to employing cue strengths in the SA support system design to mitigate information overload and wrong schema instantiation for remote engine room operators at the future SCC.

6. Conclusion

Previous incidents from the control station environments demonstrate that the SA is not limited to the physical maneuvering of an object in direct proximity but extends to knowing the status of a distant system. However, within the maritime community, it still appears that the locus of SA support systems is limited to remote navigation and visual awareness, albeit a human-centred design approach is vital to mitigate a potential risk of information overload and poor comprehension during the remote ERM. This study developed two Likert-type questionnaires comprised of five-point importance scales and surveyed 201 marine engineers to investigate the cue strengths of perception information. The Mann–Whitney U test was employed to analyse the perspective generality through two group comparisons by three categories: ship types, ranks and experience of respondents. The importance ratings reveal that cue strength differences exist in comprehending engine room machinery and offer a promising strategy for addressing information overload and wrong schema activation. Ship types and rank comparisons show that the significant disagreement rates on the importance rating are between 5 and 12% of total perception elements. For experience comparisons, the respondents are divided by experience thresholds from 1.5 to 14 years with a 0.5-year increment. The result indicates that the disagreement rate between groups reduces along with the increase of year thresholds. It is likely because the vessel rotation for engineers is typical, enabling them to accumulate vast knowledge across different types of ships. For the future ERM survey, a small number of highly experienced or a large number of less experienced engineers can be involved, as their perspective differences become almost negligible. One limitation of this study is that the evaluation criteria for importance rating can vary individually. One may consider a particular piece of information important because of the commonality, criticality and range of monitoring events that it can reflect. The following research will investigate the effect of key transparent information on SA through simulation experiments.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The participants of this study did not give written consent for their data to be shared publicly, so due to the sensitive nature of the research supporting data is not available.

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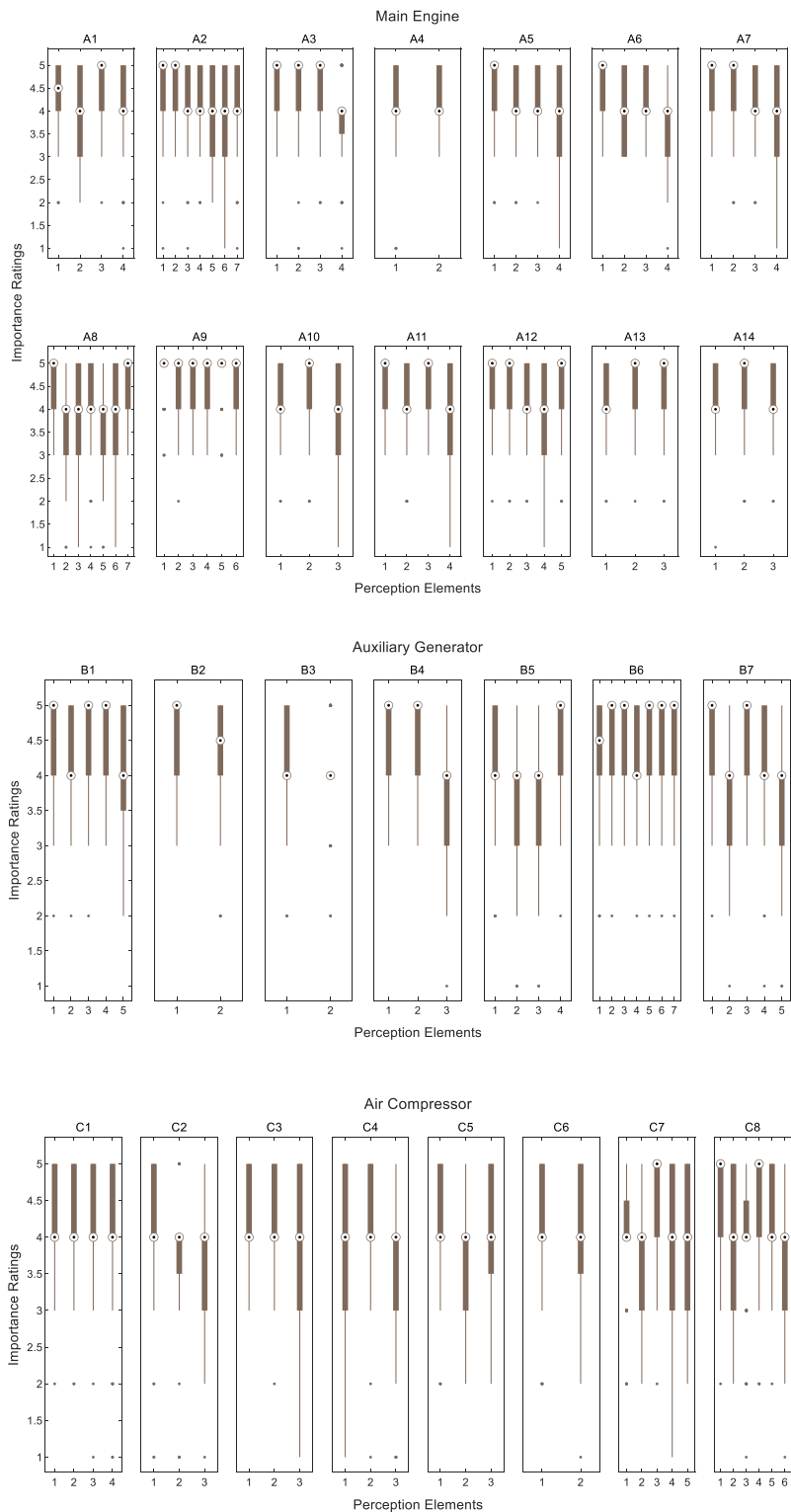
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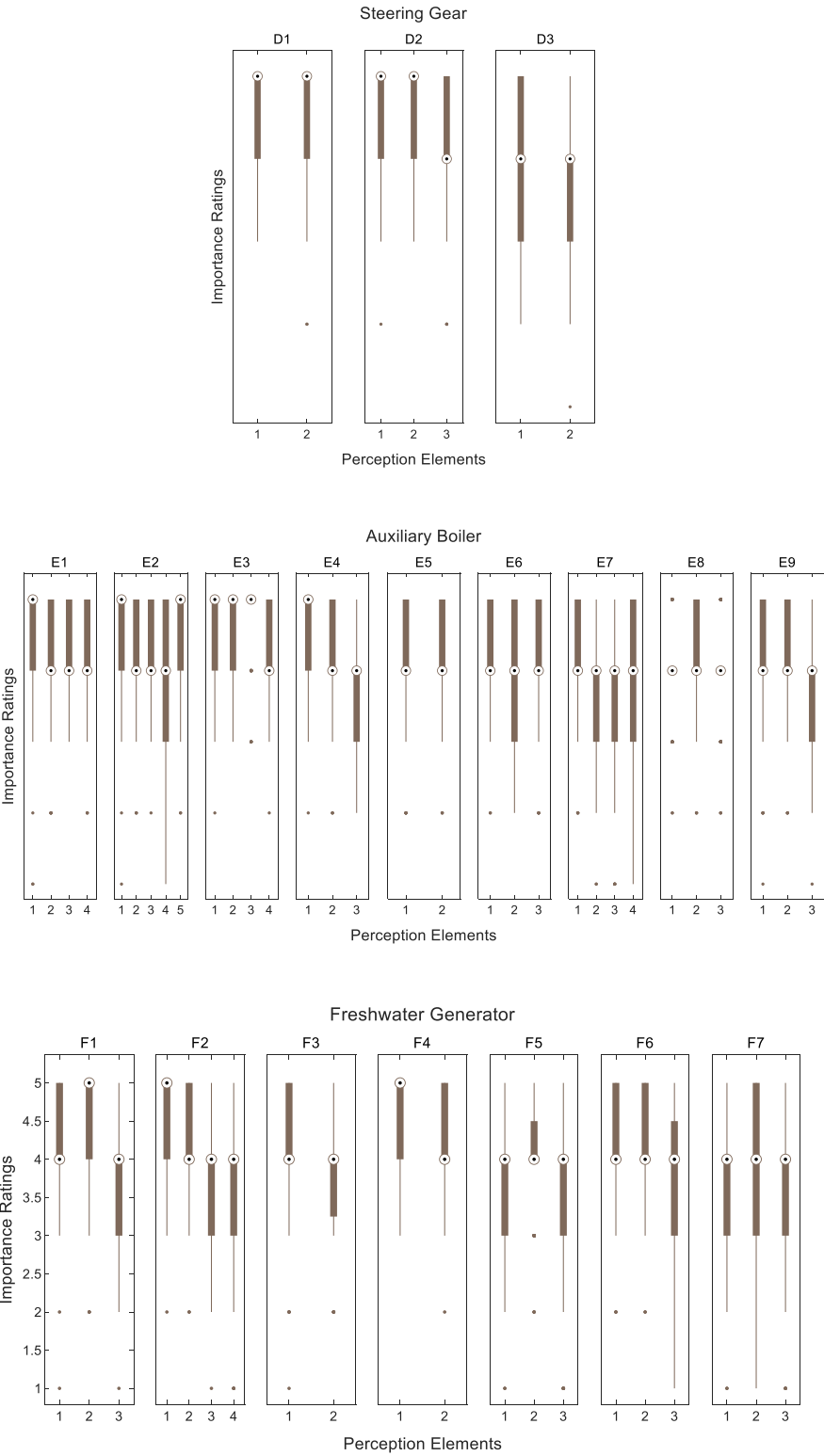
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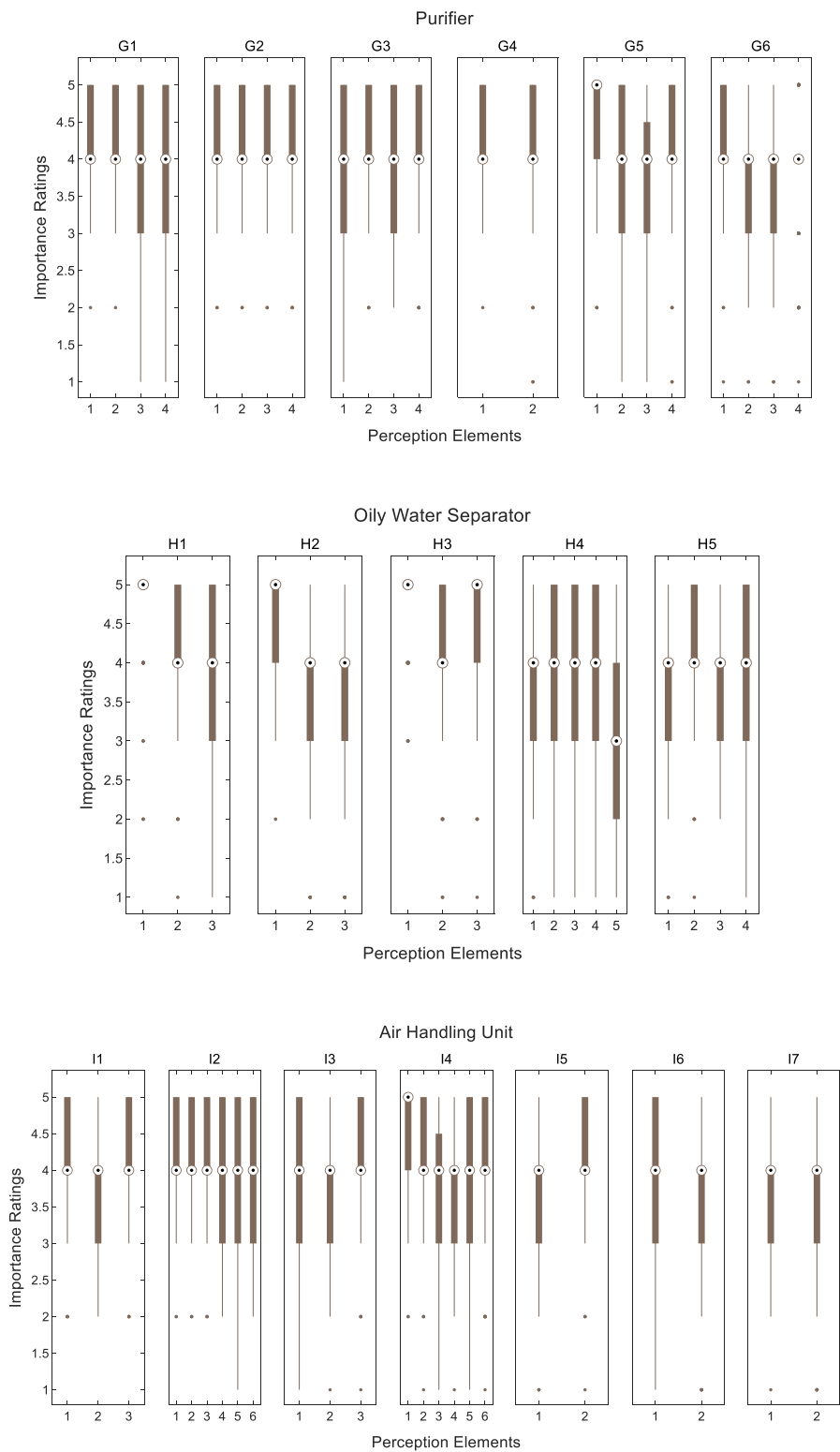
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Appendices

Appendix A. Central transparency for SA information requirement







Appendix B. Importance ratings for perception information in median

A. Main Engine (2 Stroke)			
A1. Status of power generation			
1. Indicative power	[4.5]		
2. Brake power	[4]		
3. Engine RPM	[5]		
4. Target engine RPM	[4]		
<i>*RPM: Revolution per minute</i>			
A2. Status of engine load			
1. Current RPM	[5]		
2. Current fuel index	[5]		
3. Current shaft torque	[4]		
4. Fuel index and shaft torque at NCR	[4]		
5. Fuel index and shaft torque at MCR	[4]		
6. Fuel index and shaft torque at OR	[4]		
7. Torque rich zone	[4]		
<i>*NCR: Normal continous rating</i>			
<i>*MCR: Maximum continous rating</i>			
<i>*OR: Overload rating</i>			
A3. Adequacy of combustion			
1. Deviations of each cylinder's P-max from average P-max	[5]		
2. Deviations of each cylinder's P-comp from average P-comp	[5]		
3. Deviations of each cylinder's EG temp from average EG temp	[5]		
4. Colours of EG	[4]		
<i>*P-max: Maximum combustion pressure</i>			
<i>*P-comp: Compression pressure</i>			
<i>*EG: Exhaust gas</i>			
<i>*Temp: Temperature</i>			
A4. Adequacy of auxiliary blowers			
1. Auto stop and start pressure	[4]		
2. Scavenging air tank pressure	[4]		
A5. Adequacy of turbocharger			
1. Turbocharger RPM	[5]		
2. Inlet and outlet temp	[4]		
3. Turbine differential pressure	[4]		
4. Blower outlet pressure	[4]		
A6. Adequacy of air cooler			
1. Scavenging air inlet and outlet temp	[5]		
2. LTCW in and out temp	[4]		
3. Scavenging air differential pressure	[4]		
4. Drain quantity	[4]		
A7. Adequacy of fuel supply			
1. Engine fuel inlet pressure	[5]		
2. Engine fuel inlet temp	[5]		
3. Fuel filter differential pressure	[4]		
4. Leakage tank level	[4]		
A8. Adequacy of fuel consumption rate			
1. Daily fuel consumption rate	[5]		
2. Average daily fuel consumption rate during past voyage	[4]		
3. Average daily fuel consumption rate during sea trial	[4]		
4. Ship speed	[4]		
5. Ship loading condition	[4]		
6. Slip	[4]		
7. Engine rpm	[5]		
A9. Adequacy of system lubrication			
1. LO temp at bearings, pistons, crossheads and cams	[5]		
2. LO temp at turbocharger inlet	[5]		
3. LO temp at engine outlet	[5]		
4. LO temp at engine inlet	[5]		
5. LO pressure at engine inlet	[5]		
6. LO pressure at turbocharger inlet	[5]		
<i>*LO: Lubricating oil</i>			
A10. System lubricant quantity			
1. Sump tank level	[4]		
2. Sump tank low alarm level	[5]		
3. Sump tank level rate of change	[4]		
A11. System lubricant quality			
1. Viscosity	[5]		
2. Colour	[4]		
3. LO filter differential pressure	[5]		
4. LO filter differential pressure rate of change	[4]		
A12. Adequacy of cylinder lubrication			
1. Injection timing	[5]		
2. Injection quantity	[5]		
3. Cyl LO pressure	[4]		
4. Cyl LO temp	[4]		
5. Engine RPM	[5]		
<i>*Cyl: Cylinder</i>			
A13. Adequacy of compressed air			
1. Starting air inlet pressure	[4]		
2. Control air inlet pressure	[5]		
3. Safety air inlet pressure	[5]		
A14. Adequacy of cylinder cooling			
1. HTCW engine inlet temp	[4]		
2. HTCW engine outlet temp	[5]		
3. HTCW engine inlet pressure	[4]		
<i>*HTCW: High temperature cooling water</i>			
B. Generator (4 Stroke)			
B1. Status of electricity generation			
1. Current voltage	[5]		
2. Current ampere	[4]		
3. Current frequency	[5]		
4. Current generator load [kw]	[5]		
5. Power factor	[4]		
B2. Status of generator load			
1. Current generator load [kw]	[5]		
2. Generator capacity [kw]	[4.5]		
B3. Adequacy of three-phase wiring			
1. R,S,T temp	[4]		
2. Alternator cooling air or water temp	[4]		
B4. Adequacy of alternator bearing lubrication			
1. LO temp	[5]		
2. LO level	[5]		
3. LO colour	[4]		
B5. Adequacy of fuel consumption rate			
1. Daily fuel consumption rate	[4]		
2. Average daily fuel consumption rate during past voyage	[4]		
3. Average daily fuel consumption rate during sea trial	[4]		
4. Generator load	[5]		
B6. Adequacy of system lubrication			
1. LO temp at turbocharger inlet	[4.5]		
2. LO temp at engine inlet	[5]		
3. LO temp at engine outlet	[5]		
4. LTCW temp at LO cooler inlet and outlet	[4]		
5. LO pressure at turbocharger inlet	[5]		
6. LO pressure at LO pump outlet	[5]		
7. LO pressure at LO filter inlet and outlet	[5]		
B7. System lubricant quality			
1. Viscosity	[5]		
2. Colour	[4]		
3. LO filter differential pressure	[5]		
4. LO filter differential pressure rate of change	[4]		
5. Sludge thickness in centrifugal filter at the given time	[4]		
C. Air compressor			
C1. Adequacy of compressed air production			
1. Main reservoir pressure	[4]		
2. Service reservoir pressure	[4]		
3. Emergency reservoir pressure	[4]		
4. Air compressor daily running hour	[4]		
C2. Adequacy of running hour			
1. Daily running hour	[4]		
2. Average daily running hour	[4]		
3. Tasks requiring compressed air	[4]		
C3. Adequacy of auto operation mode			
1. Main reservoir pressure	[4]		
2. Start and stop pressure of air compressors	[4]		
3. Safety valve setting pressure	[4]		

C4. Efficiency of compression		
1. 1 st stage air outlet pressure	[4]	
LP		
2. 2 nd stage air outlet pressure HP	[4]	
3. Speed of staining stop pressure from start pressure	[4]	
<i>*HP: 2nd stage air outlet pressure</i>		
<i>*LP: 1st stage air outlet pressure</i>		
C5. Efficiency of cooler		
1. Temp difference of compressor cooler LTCW side	[4]	
2. 1 st stage air outlet temp	[4]	
3. 2 nd stage air outlet temp	[4]	
<i>*LTCW: low temperature cooling water</i>		
C6. Dryness of compressed air		
1. Water drain quantity from air reservoir	[4]	
2. Water drain frequency	[4]	
C7. Oiliness of compressed air		
1. Oil drain quantity from air compressors	[4]	
2. Oil drain frequency	[4]	
3. Air compressor LO level	[5]	
4. LP	[4]	
5. HP	[4]	
C8. Adequacy of lubrication		
1. LO level	[5]	
2. LO colour	[4]	
3. Presence of form in LO	[4]	
4. Presence of water in LO	[5]	
5. LO pump outlet pressure	[4]	
6. Oil drain quantity from air compressors	[4]	
D. Steering Gear		
D1. Adequacy of rudder operation		
1. Deviation of rudder angle from command angle	[5]	
2. Rudder response speed	[5]	
D2. Adequacy of SG hydraulic system		
1. Hydraulic tank level	[5]	
2. Hydraulic oil pressure	[5]	
3. Hydraulic oil temp	[4]	
D3. Adequacy of SG injection		
1. Grease injection timer setting	[4]	
2. Manual grease injection interval	[4]	
E. Auxiliary Boiler		
E1. Adequacy of steam production		
1. Boiler auto stop and start pressure	[5]	
2. Target steam pressure	[4]	
3. Current steam pressure	[4]	
4. Boiler load	[4]	
E2. Adequacy of boiler auto operation		
1. Auto stop and start pressure	[5]	
2. Target steam pressure	[4]	
3. Current steam pressure	[4]	
4. Dump steam setting	[4]	
pressure		
5. Safety valve setting pressure	[5]	
E3. Consistency of boiler water level		
1. Deviation of current boiler water level from setting level	[5]	
2. Low alarm level	[5]	
3. Low low alarm level	[5]	
4. High alarm level	[4]	
E4. Adequacy of fuel supply		
1. Fuel pressure at nozzle	[5]	
2. Fuel temp at nozzle	[4]	
3. Ampere of rotary cup motor	[4]	
E5. Adequacy of combustion air		
1. Fan motor ampere	[4]	
2. Fan damper position	[4]	
E6. Adequacy of combustion status		
1. Colours of EG	[4]	
2. EG temp	[4]	
3. Flame colour	[4]	
E7. Adequacy of fuel consumption rate		
1. Daily fuel consumption rate	[4]	
2. Average fuel consumption rate during past voyage	[4]	
3. Average fuel consumption rate during sea trial	[4]	
4. Boiler load	[4]	
E8. Efficiency of boiler heat exchange		
1. Combustion chamber temp	[4]	
2. Boiler outlet EG temp	[4]	
3. Boiler load	[4]	
E9. Efficiency of economizer heat exchange		
1. Differential pressure (exhaust side)	[4]	
2. Temp difference (exhaust side)	[4]	
3. Temp difference (water side)	[4]	
F. Freshwater Generator		
F1. Adequacy of freshwater production		
1. Freshwater production rate	[4]	
2. Salinity level	[5]	
3. Seawater temp	[4]	
F2. Adequacy of vacuum forming		
1. Shell vacuum pressure	[5]	
2. Shell temp	[4]	
3. Seawater temp	[4]	
4. Speed of attaining target vacuum pressure	[4]	
F3. Efficiency of heat exchange		
1. Temp difference (evaporator JKCW side)	[4]	
2. Temp difference (condenser CSW side)	[4]	
F4. Adequacy of demister		
1. Salinity level	[5]	
2. Freshwater production rate	[4]	
F5. Adequacy of distilled water transfer		
1. Deviation of distilled water pressure from target discharge pressure	[4]	
2. Distilled water flowrate	[4]	
3. Corresponding freshwater tank level to freshwater generator	[4]	
F6. Efficiency of ejector		
1. Ejector pressure	[4]	
2. Shell vacuum pressure	[4]	
3. Brine level	[4]	
F7. Adequacy of chemical dosage		
1. Feedwater flowrate	[4]	
2. Chemical dosage rate	[4]	
3. Chemical tank level	[4]	
G. Purifier		
G1. Status of oil purification		
1. Oil feed rate	[4]	
2. Purifier discharge pressure	[4]	
3. Service tank level	[4]	
4. Settling tank level	[4]	
G2. Stability of fuel discharge pressure		
1. Purifier discharge pressure	[4]	
2. Purifier fuel outlet temp	[4]	
3. Discharge deviation alarm pressure	[4]	
4. Discharge low alarm pressure	[4]	
G3. Adequacy of sludge discharge		
1. Partial discharge rate at the given time	[4]	
2. Total discharge timer setting	[4]	
3. Sludge tank level	[4]	
4. Operating water pressure	[4]	
G4. Adequacy of purifier motor		
1. Motor ampere	[4]	
2. Purifier RPM	[4]	
G5. Adequacy of lubrication		
1. LO level	[5]	
2. LO colour	[4]	
3. Presence of foam in LO	[4]	
4. Presence of water in LO	[4]	
G6. Adequacy of purifier heater		
1. Deviation of fuel outlet temp from target fuel temp	[4]	
2. Steam inlet pressure	[4]	
3. Steam outlet pressure	[4]	
4. On-off percentage of steam regulating valve	[4]	
H. Oily Water Separator		
H1. Adequacy of bilge discharge		
1. Bilge PPM	[5]	
2. Discharge rate	[4]	
3. Discharge pressure	[4]	
H2. Adequacy of bilge treatment		
1. Differential pressure of	[5]	



OWS filters	
2. Bilge heater bilge outlet temp	[4]
3. Bilge heater target temp	[4]

H3. Adequacy of pollution prevention sequence

1. Bilge PPM	[5]
2. Oily content in OWS tanks	[4]
3. On-off position of solenoid valves	[5]

H4. Adequacy of bilge holding tank

1. Tank capacity	[4]
2. Tank level	[4]
3. Bilge accumulation rate	[4]
4. Oily content in the tank	[4]
5. Bilge temp	[3]

H5. Adequacy of bilge well

1. Well capacity	[4]
2. Well level	[4]
3. Bilge accumulation rate	[4]
4. Oily content in the tank	[4]

I. Air Handling Unit

I1. Adequacy of accommodation cooling

1. Deviation of evaporator air outlet temp from evaporator target air outlet temp	[4]
2. Atmospheric temp	[4]
3. Accommodation temp	[4]

I2. Adequacy of AHU compressor auto operation mode

1. Auto stop and start pressure	[4]
2. Trip pressure	[4]
3. AHU LP	[4]
4. AHU HP	[4]
5. Compressor running stage	[4]
6. Compressor ampere	[4]

*AHU: Air handling unit

*LP: Compressor refrigerant inlet pressure

*HP: Compressor refrigerant outlet pressure

I3. Adequacy of AHU condenser

1. Temp difference (cooling water side)	[4]
2. Pressure difference (cooling water side)	[4]
3. Refrigerant level	[4]

I4. Adequacy of lubrication

1. LO level	[5]
2. LO pump outlet pressure	[4]
3. LO temp	[4]
4. LO colour	[4]
5. Presence of foam in LO	[4]
6. Presence of water in LO	[4]

I5. AHU fan

1. Fan RPM	[4]
2. Motor ampere	[4]

I6. Adequacy of humidification

1. Deviation of accommodation humidity	[4]
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from humidifier target humidity

2. Humidifier steam or freshwater consumption rate	[4]
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I7. Adequacy of heating

1. Deviation of accommodation temp from heater target temp	[4]
2. Steam consumption rate	[4]